

UPSC Prelims

Science & Tech

Class No. 11

Defence Technology

Goldilocks zone

The 'Goldilocks Zone,' or habitable zone, is the range of distance with the right temperatures for water to remain liquid.

Discoveries in the Goldilocks Zone, like Earth-size planet Kepler-186f, are what scientists hope will lead us to water—and one day life.

Ex : our earth is in the sun's goldilock's zone

Greased Lightning 10 : NASA

The GL-10 Greased Lightning is a hybrid diesel-electric tilt wing unmanned aircraft.

It is a battery-powered plane that has 10 engines and can take off like a helicopter and fly efficiently like an aircraft.

One other advantage to the GL-10 besides its versatile vertical take off and landing ability is its noise or lack of it.

Ice Cube Neutrino Observatory

The IceCube Neutrino Observatory is an array of 5,160 optical sensors buried in a cubic kilometer of ice beneath the South Pole.

- experiment detect light that is produced when neutrinos interact with a nucleus in the surrounding medium.
- neutrinos are weakly interacting, nearly massless fundamental particles might be produced (according to some theories) when particles of dark matter annihilate with each other. If these theories are true, neutrino detectors—like the IceCube Neutrino Observatory at the South Pole - can look for excess neutrinos coming from known dark matter hotspots.

Missile Tech

- A missile is a guided airborne ranged weapon capable of self-propelled flight usually by a jet engine or rocket motor.
- Missiles have five system components: targeting, guidance system, flight system, engine and warhead.
- A missile needs a flight system. Guided missile is a missile whose course may be altered during flight.
- Range: Land based missiles have longer range than air-based version. Submarine versions have least range.
- The Mysorean rockets were the first iron-cased rockets in world that were successfully deployed for military use. Hyder Ali and Tipu Sultan used the rockets effectively against the British East India Company during the 1780s and 1790s.

Missile Classification

Range

- a.) Short Range (≤ 500) : prithvi
- b.) Medium " (≤ 1500)
- c.) Intermediate (≤ 5000 km) : Agni
- d.) Long Range (> 5000 km) : ICBM. (Agni-IV)

Missile Classification

Based on launch site
and target

- a.) Surface to Surface Missile (prithvi)
- b.) Surface to Air (Aakash) (Trishul) (Barak).
 - not used now
- c.) Air to Air (Astra)
 - used by navy built in israel.

Missile Classification

Speed

- a) Subsonic (< 1 Mach) < 332 m/s : Nirbhaya
- b) Supersonic (1-5 Mach) : Brahmos
- c) Hypersonic (> 5 Mach) → proposed: Shaurya
→ (trying to upgrade Brahmos)

Missile Classification

Trajectory followed by missile :

- a) Ballistic
- b) Cruise

Ballistic v/s Cruise Missile

Ballistic : Predetermined Target

Most of the flight is unpowered and unguided after initial power by rocket.

Long Range : Outside atmosphere

ICBM : sub orbital trajectory

Tactical Ballistic Missile : Battlefield Range Ballistic Missile. Short range use <300 km

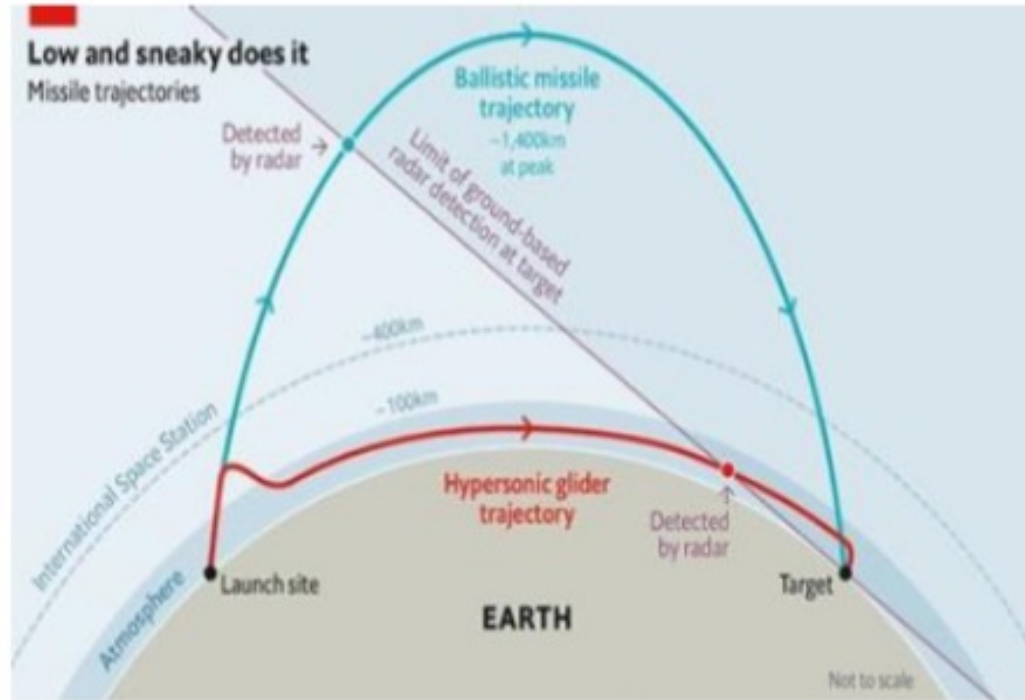
Cruise : Full flight is powered hence it is a self propelled guided missile

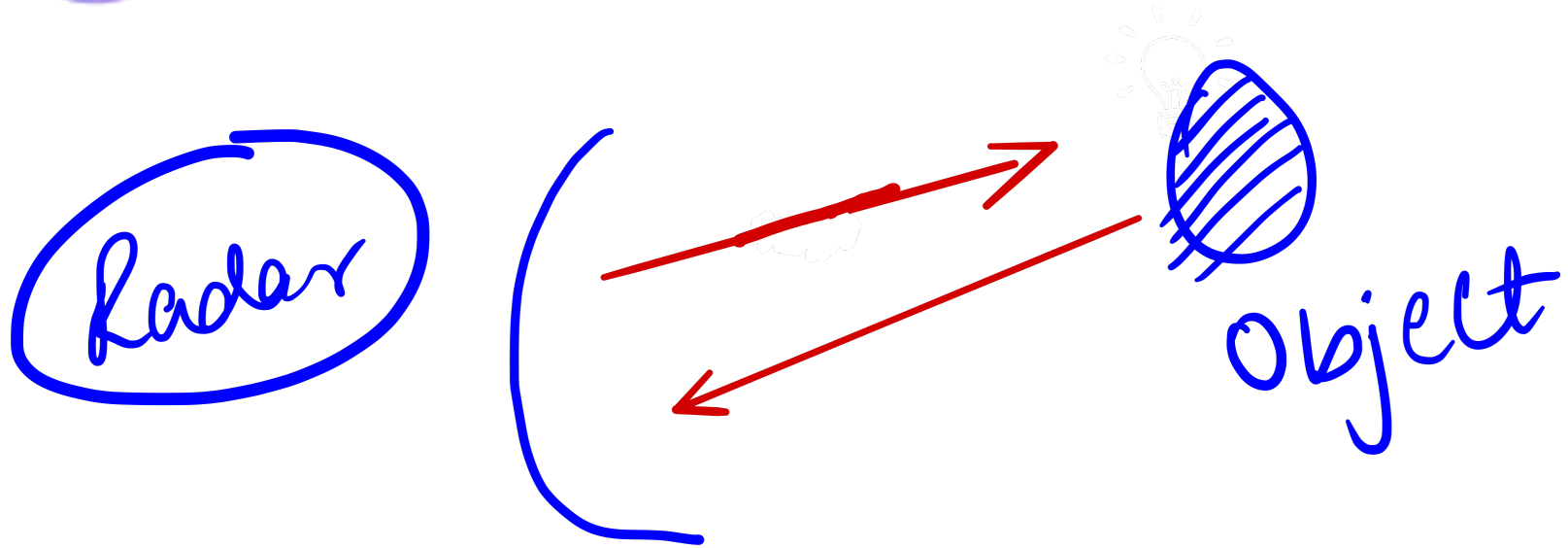
Within the earth's atmosphere : Jet Engine Technology

used against terrestrial targets. It flies the major portion of its flight path at approximately constant speed.

Examples of cruise missiles : Brahmos, Harpoon (USA), Exocet (France)

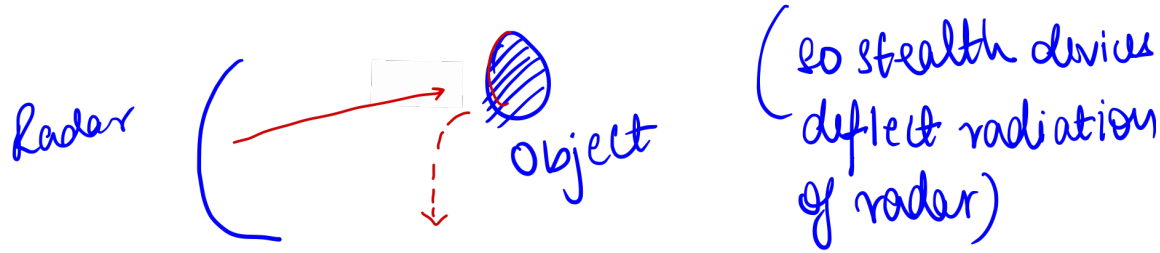
Ballistic	Cruise
Depends on Gravity to hit the target	Does not depend on gravity
2 phases : exo and endoatmospheric	Remains endoatmospheric
Advantages : Range is more, Saves Fuel	Advantages: Precise and High speed, Difficult to intercept by Radar.
Disadvantages : Can be intercepted by radar / Anti Ballistic Missile.	Disadvantages : Requires continuous guidance
Better for Fixed Targets but not for mobile ones	More fuel is needed
	Only short and medium Range





Stealth Technology

① designed to deflect radiation



② Surface coating (absorbs radiation)
→ use of material science

Classification : Launch Modes

- Surface to surface
- Surface-to-Air Missile
- Air-to-Air Missile
- Air-to-Surface Missile
- Sea-to-Sea Missile
- Torpedo
- Anti-Tank Missile

Ballistic Missiles

ICBM : intercontinental : $>5000\text{km}$

IRBM : intermediate range : $3000 - 5000\text{km}$

MRBM : Medium Range Ballistic Missile : $1000-3000\text{km}$

SRBM : Short Range Ballistic Missile : $<1000\text{km}$

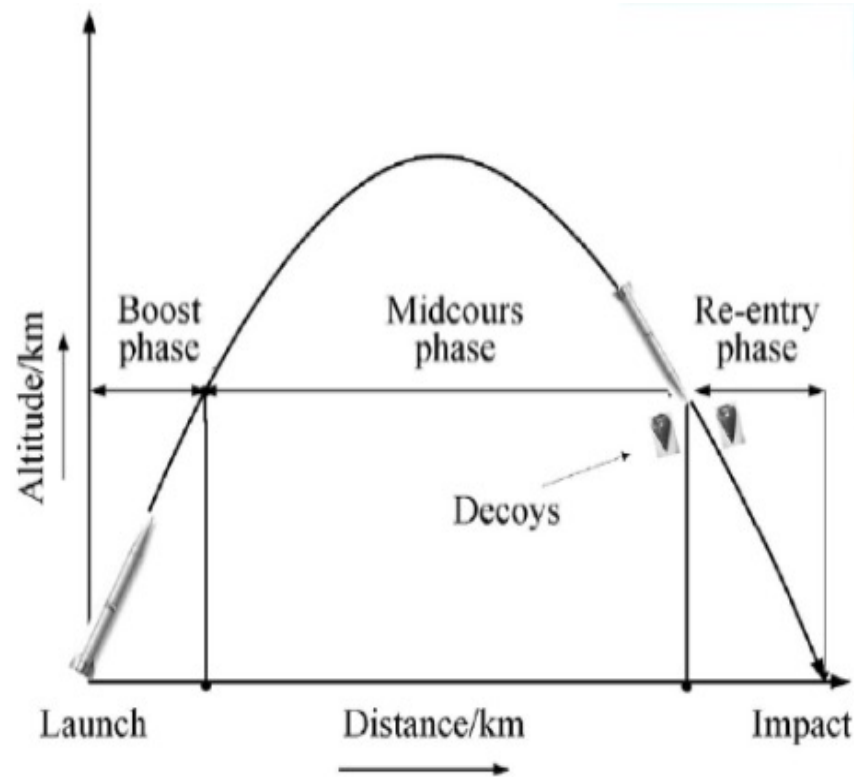
Tactical : For use in the battlefield

ICBM

- Inter continental Ballistic Missile
- It is a missile with a minimum range of 5,500 kilometres primarily designed for nuclear weapons delivery.
- India, Russia, the United States, China, the United Kingdom and France, Israel and North Korea are the only countries that have operational ICBMs.
- Modern designs support Multiple Independently Targetable Reentry Vehicles (MIRVs).
- A MIRV is an exo-atmospheric ballistic missile payload containing several warheads. Each one of them is capable of being aimed to hit a different target.

Test Question

- Which are the Non-NPT countries of the world?



Hypersonic Missiles

- DRDO has successfully tested a Hypersonic Technology Demonstrated Vehicle (HSTDV) in September 2020, and demonstrated its hypersonic air-breathing scramjet technology.
- The significance of developing the technology has come in the backdrop of development strides made by China, Russia and US in the domain.
- Hypersonic defines speed of about Mach 5 or at least at 1.6 km per second.
- Ballistic missiles, follow a fixed trajectory and travel outside the atmosphere to re-enter only near impact.
- On the contrary, hypersonic weapons travel within the atmosphere and can manoeuvre midway which combined with their high speeds makes their detection and interception extremely difficult.
- This means that radars and air defences cannot detect them till they are very close and little time to react.

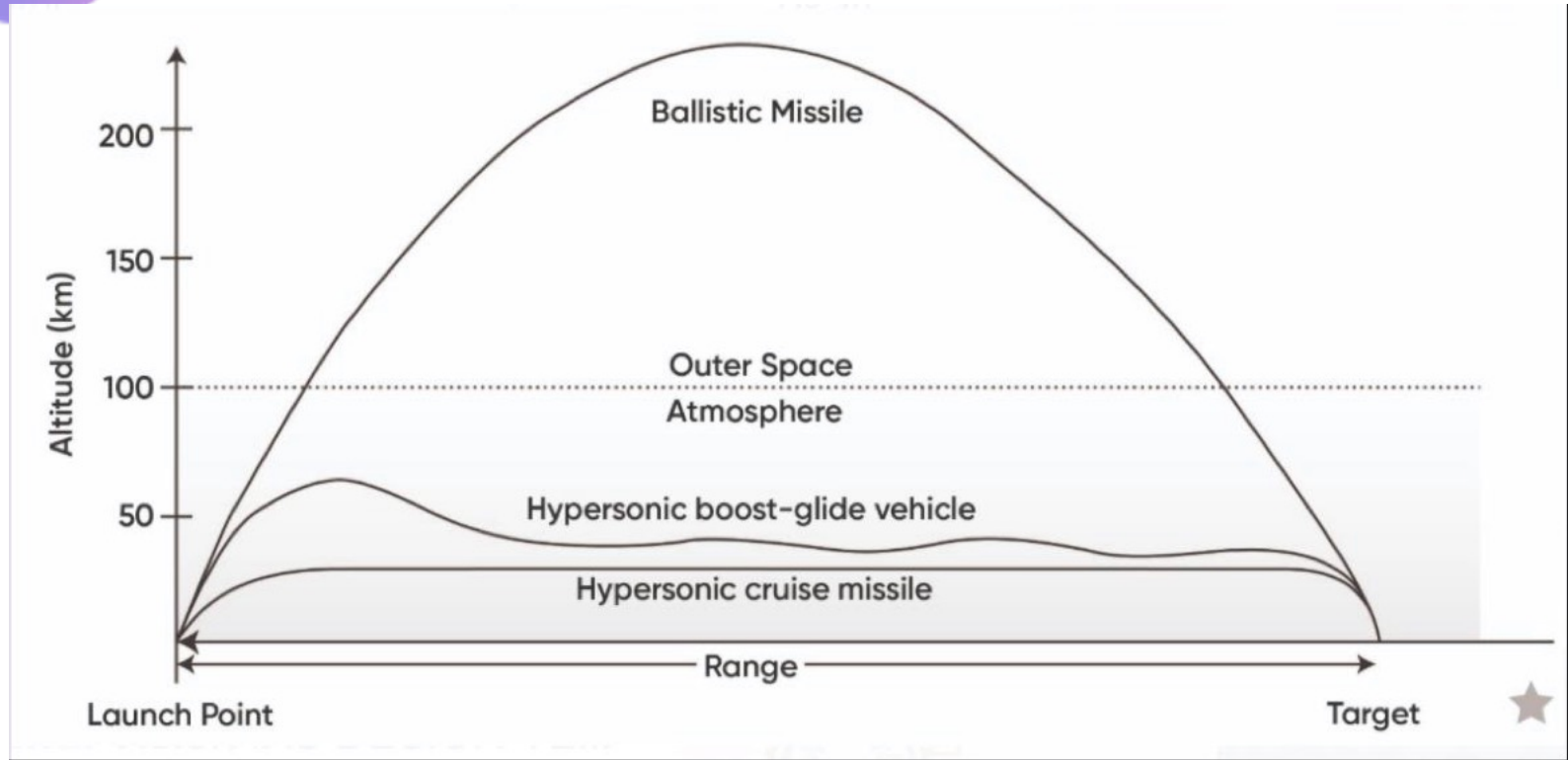
Hypersonic Missiles

2 Types of hypersonic platforms:

- o Hypersonic cruise missiles: Use rocket or jet propellant through their flight and are regarded as being faster versions of existing cruise missiles.

- o Hypersonic Glide Vehicle (HGV):

These missiles first go up into the atmosphere on a conventional rocket before being launched towards their target.



Hypersonic weapons

Considered the next generation of arms with conventional or nuclear warheads that are hard to detect and can travel more than five times the speed of sound

TRAJECTORY AND DETECTION

HYPERSONIC GLIDE VEHICLE

Speed: at least Mach 5

Launched from a rocket. Hard to detect, track and intercept

Hypersonic cruise missiles are powered by "scramjets", maneuvered at lower altitude toward target

Detected by radar

Limit of ground-based radar detection at target

Atmosphere (100 km)

CONVENTIONAL BALLISTIC MISSILE

Between Mach 1 and Mach 5

Easier to predict and intercept

Detected by radar

Launch

Target

Mach Number

Mach number : Speed relative to speed of sound.

Velocity Zone	Mach Number
Subsonic	Mach < 1
Trans-sonic	Mach ~ 1
Supersonic	Mach > 1.0
Hypersonic	Mach > 5.0

Hypersonic Weapons

- In August, China reportedly tested a nuclear-capable hypersonic glide vehicle that circled the globe before speeding towards its target.
- Chinese appear to have combined a Fractional Orbital Bombardment System, or FOBS, with a hypersonic weapon. FOBS sends missiles through a partial orbit around the earth to strike targets from an unexpected direction.
- According to the reports, weapon could, in theory, fly over the South Pole. That would pose a big challenge for the US military because its missile defence systems are focused on the northern polar route.
- This test was not in isolation. China is building up his strategic forces across the board, quadrupling the size of the arsenal and building new siloed intercontinental ballistic missiles (ICBMs), new mobile ICBMs, new nuclear bombers, new nuclear-armed submarines, new missile defences, and new hypersonic missiles.



Intercontinental Ballistic Missiles (ICBMs)	Hypersonic Cruise Missiles (HCMs)
• Traditional ballistic missiles and ICBMs arc along a predictable ballistic path, like a bullet, and lack the element of surprise. • Assessment by the Nuclear Threat Initiative premises that the President of the US has just 2-4 minutes to render a decision on intercepting incoming ballistic missiles fired by China/Russia and launch its own missiles against the latter.	• Hypersonic missile packs the manoeuvrability of much slower subsonic cruise missiles and the hypersonic speed greater than or equal to ICBMs, making it harder to track and defend against. • The detection-interception time in case of HCMs and HGVs would be far less. Besides, HGVs do not rise above 100-110 kms altitude and HCMs fly even lower, at about 20-30 kms altitude. ○ Their flatter trajectories vis-à-vis ICBMs, along with the Earth's curvature, further complicates the dynamics of detecting-intercepting.

Missile Defence System

Missile defense systems are intended to shield against incoming missiles, such as intercontinental ballistic missiles (ICBMs) or other ballistic missiles.

The United States, Russia, India, France, Israel, Italy, United Kingdom, China and Iran have developed missile defense systems.

Think : Easier to defend against which missile :

Ballistic or Cruise?

Supersonic or subsonic or Hypersonic?

Missile Defence System : Various Countries

France UK and Italy : PAAMS (principle anti air missile system). Also known as Sea Viper in the UK.

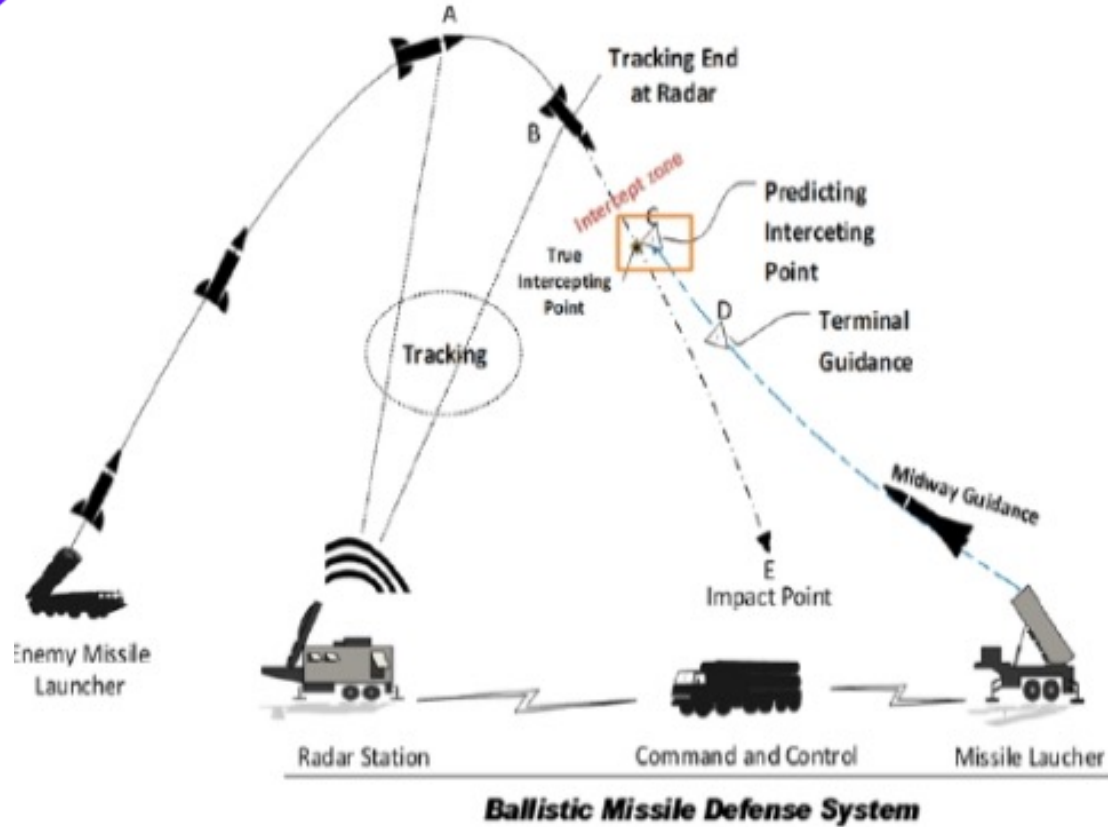
France and Italy also have Aster Missile System

Israel : Arrow missiles, Iron Dome, Iron Beam, Barak -8 Missiles

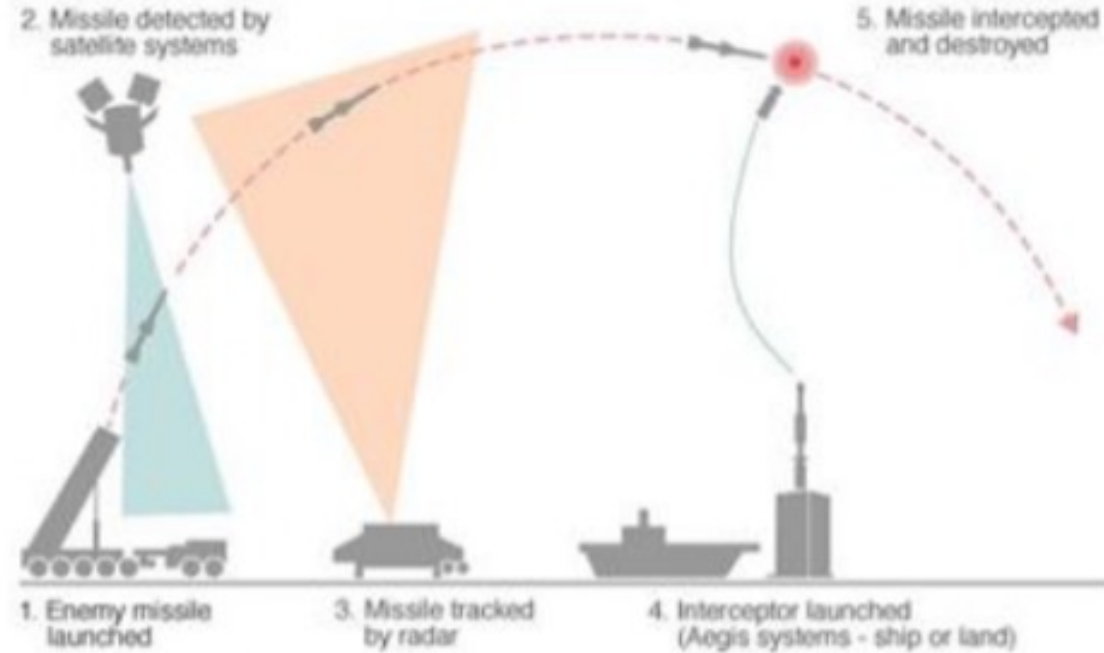
USA : Terminal High Altitude Aerial Defence (THAAD)

China : KT and HQ series of Missiles

Russia : A135, A35, S400, S300



How the missile defence system works



Source: Missile Defence Agency

India's Ballistic Missile Defence

It is an initiative to develop and deploy a multi-layered ballistic missile defense system to protect India from ballistic missile attacks.

Why? : Hostile Neighbors

It can intercept an incoming ballistic missile launched from 5,000 km.

It is a double-tiered system consisting of two interceptor missiles, namely PAD and AAD.

PAD is for high-altitude interception, and the Advanced Air Defence (AAD) Missile for lower altitude interception.

Pradhyuman : exoatmospheric

Ashwin : Endoatmospheric

Prithvi / Pradhyumna Air defence

The PAD is an anti-ballistic missile developed to intercept incoming ballistic missiles outside the atmosphere (exo- atmospheric, 80 km altitude).

It is based on the Prithvi missile, which is a tactical surface-to-surface short-range ballistic missile (SRBM)

PAD is a two-stage missile. The first stage is solid fuelled, while the second stage is liquid fuelled.

Guidance is provided by internal navigation system with active radar homing in the terminal phase.

PAD is fast enough to hit medium-range ballistic missiles and intermediate-range ballistic missiles.

Advanced Air Defence

AAD is an anti-ballistic missile designed to intercept incoming ballistic missiles in the endo-atmosphere at an altitude of 40 km.

AAD is a single stage, solid-fueled missile.

Guidance is similar to PAD

Difference: Indigenous RFID seeker

S 400

S 400 is Russia's 4th generation of long- range surface-to-air missile system.

The evolved version of the S-400 system is the S-500 Prometheus.

India has already procured a squadron of S- 300V systems which are in use as an anti- tactical ballistic missile screen'.

As per SIPRI : one of the most advanced air defense systems in the world

Stockholm International Peace Research Institute

capable of firing 3 types of missiles to create a layered defense.

anti-aircraft missile system : designed to destroy aircraft, cruise, and ballistic missiles. It can also be used against ground targets.

S 400

The S-400 is able to intercept cruise missiles at a range of about 40 km due to their low- altitude flight paths.

The anti-ballistic missile (ABM) capabilities of the S-400 system are near the maximum allowed under the Anti-Ballistic Missile Treaty.

Can engage all types of targets : UAV / Ballistic / Cruise

Range : 400km

Equipped with four different missiles, it can engage enemy aircraft, ballistic missiles, and Airborne Warning And Control System (AWACS) planes at 400km, 250km, medium-range 120km and short-range 40km.

It has the capability to engage 80 targets at one time with a response time of 9-10 seconds.

ABM treaty

It was an arms control treaty between the United States and the Soviet Union on the limitation of the anti-ballistic missile (ABM) systems used in defending areas against ballistic missile-delivered nuclear weapons. Under the terms of the treaty, each party was limited to two ABM complexes, each of which was to be limited to 100 anti-ballistic missiles.

Signed in 1972

In 2002 : The US withdrew

Cruise Missile Detection

Low trajectory : difficult to intercept via conventional methods

AAD : can intercept cruise missiles

Airborne early warning and control system :
AEW&CS



Defence Related Facts : Indian Missiles

Missile	Type and Details	From - to	Range (km)	Remarks
Prithvi I	SRBM	S2S	150	India's first indigenous missile, Nuclear capable, not in service
Prithvi II	SRBM	S2S	350	In service
Prithvi III	SRBM	S2S	600	In service
Dhanush	Naval variant of Prithvi II	Sea to Surface	350	In service
Agni I	MRBM, Single Stage	S2S	1250	In service
Agni II	IRBM, Two Stage	S2S	2000	In service
Agni III	IRBM, Two Stage	S2S	3000	In service
Agni IV	IRBM, Two Stage	S2S	4000	In service
Agni V	ICMB, 3 stage	S2S	5000	In service
Agni VI	ICBM, 4 stage	S2S	5500	In development
Agni P (prime)	maneuverable reentry vehicle, can be used as ASBM	S2S	Details pending	Tested in 2021
Trishul	Navy, short range	SAM	9km	Not in Service
Akash	Supersonic, Medium Range	SAM	30km	Recently, export authorised to friendly nations

Missile	Type and Details	From - to	Range (km)	Remarks
Astra	Active Radar Homing, beyond visual range	A2A	80	First A2A by DRDO, Mach 4.5 speed, Solid fuel, has radar transceiver
Shaurya	Canister based, Medium Range, Hypersonic, tactical	S2S	700-1900	In service, land based parallel of submarine launched k-15
Prahaar	Tactical Ballistic Missile, QR, Omnidirectional Warhead	S2S	150	In service, meant to replace Prithvi 1
K4	SLBM, intermediate range, in development	Underwater to Surface	3500	Named after Dr. Kalam, Nuclear triad and second strike capability
Sagarika (k-15)	SLBM	Under water to surface	700	SLBM version of Shaurya, to be integrated with INS arihant
Barak 1	Short range SAM, ship defense	Ship to air, surface	12km	Built with Israel, to be integrated with Shivalik Class frigates
Barak 8	SRSAM, ship defense	Ship to air, ship to surface	90km	Israel, to be integrated with Kolkata class destroyers.
Nirbhay	Stealth, subsonic, cruise, nuclear capable	Land, naval, air	1000km	In service, no MTCR issues

Brahmos

It is developed jointly by India and Russia. Named after Brahmaputra and Moskva rivers.

can be launched from Land, Air, Submarine and Ships.

is the world's fastest anti-ship cruise missile.

2 stage : first solid then liquid

As India became a member of MTCR in 2016, range was increased to 600km

Now the range of new land-attack version is extended to 400 km from 290 km. Speed is maintained at 2.8 Mach : supersonic



BRAHMOS

- BrahMos is a universal long range supersonic cruise missile system that can be launched from land, sea and air against surface and sea-based targets.
 - BrahMos is a collaboration between India (Defence Research and Development Organisation (DRDO) and Russia (NPO Mashinostroyeniya).
 - Name represents Brahmaputra and Moskva rivers

BRAHMOS

- A two-stage missile stage
- Fire and forget principle
- The range of the missile was originally capped at 290 km as per obligations of the Missile Technology
- Control Regime (MTCR)
- Following India's entry in MTCR in 2016, it was decided to extend the range to 450 km and to 600 km at a later stage.

Other similar missile systems

- Chinese HD-1 supersonic missile
- Tomahawk (used by U.S and its allies)
- Israel's Sea Breaker
- P-800 Oniks/Yakhont

Other Missiles

Python 5 : 5th Gen AAM, Israeli origin, integrate with Tejas, Beyond Visual Range, radar guided

Avangard : Russia deployed first hypersonic long range Nuclear-capable missiles, SCRAMJET engine

Agni Prime

DRDO successfully tested the new missile Agni Prime, advanced version of the 'Agni-1' missile.

Agni Prime' is a short-range (surface to a surface) ballistic missile that will have a range of 1000 km to 1500 km and will have advanced features in agility and road mobility.

It can carry a payload of around 1,000 Kg or a nuclear warhead.

It is a two-stage canisterised solid propellant ballistic missile with dual redundant navigation and guidance system.

Agni Prime

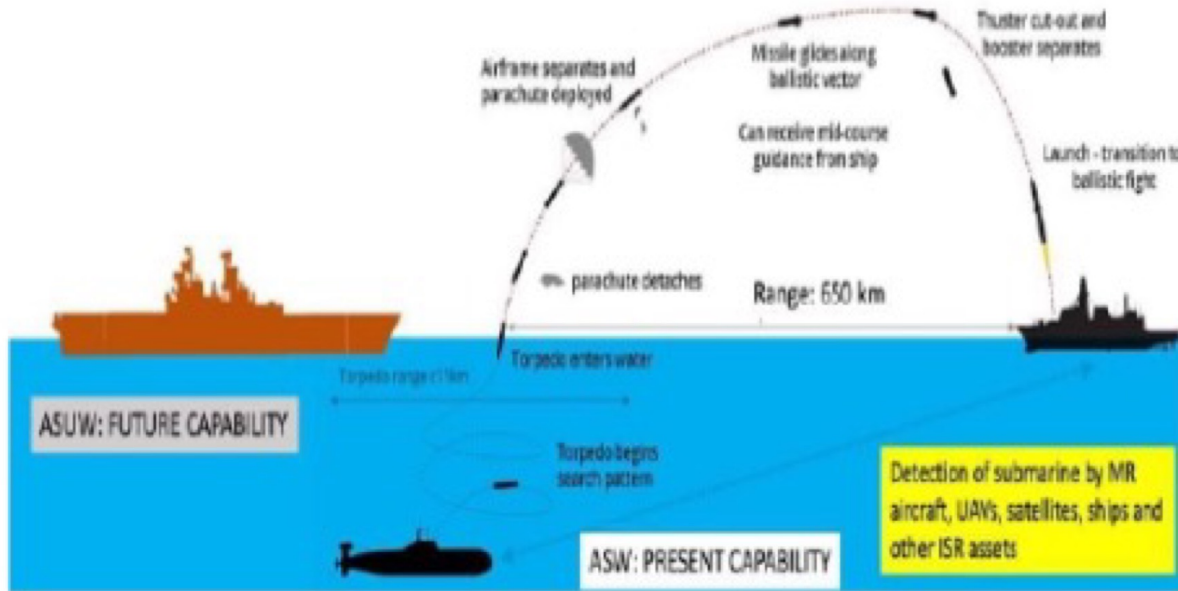
o Since the missile is canisterised, it can be launched from rail and road and stored for a longer period and transported all across the country as per operational requirements.

The double stage missile will be lighter and much sleeker than its predecessor 'Agni-1'.

Agni-1 is a short-range (surface to a surface) nuclear capable ballistic missile

Supersonic Missile Assisted Torpedo

- Recently, DRDO tested a long-range SMART from Wheeler Island in Odisha.
- SMART is a next-generation missile-based standoff torpedo delivery system.
- It has been designed to enhance anti-submarine warfare capability far beyond the conventional range of the torpedo.



Looking Ahead

India's progress towards developing hypersonic technology.

- **Brahmos II:** India has collaborated with Russia on the development of Brahmos II, a Mach 7 hypersonic cruise missile. It is scheduled to achieve initial operational capability between 2025 and 2028.
- **Mission Shakti:** To protect its space assets, India has already proved its capabilities through the test of indigenously developed anti-satellite (ASAT) missile technology.

Looking Ahead

- HSTDV: India is also developing an indigenous, dual capable hypersonic cruise missile as part of its Hypersonic (HSTDV) and successfully tested a mach 6 scramjet in June 2019 and September 2020.
- HWT test facility: An advanced Hypersonic Wind Tunnel (HWT) test facility of the DRDO was inaugurated last year in Hyderabad. It is a pressure vacuum-driven, enclosed jet facility that simulates Mach 5 to 12.

Missiles in News

- Agni 3 : IRBM
- S- 400
- Anti Tank Guided Missile : laser guided, placed on MBT arjun. Employs tandem high explosive anti tank (HEAT) warhead to counter explosive reactive armor, multi platform launch capable.
- Man Portable Anti Tank Guided Missile : indigenously developed, low weight, man portable launcher with fire and forget capability. Has miniaturized infrared seeker for on board control and guidance.
- Vertical Launch Short Range Surface to Air Missile : indigenous, ship borne, aerial threats at close range. Based on design of Astra missile (Beyond visual range air to air missile)
- QRSAM : Quick reaction SAM, indigenously designed, protective shield to moving armoured columns of the army from aerial attacks. Range : 25 – 30km

Missiles in News

- Very Short Range Air Defence System (VSHORADS) : Man portable air defence system
- HELINA : helicopter launched anti tank guided missile, Helicopter launched NAG, 3rd generation fire and forget missile mounted on ALH : All weather day and night capability
- Astra Mk-1 : BVR Air to Air missile
- Pralay Missile : Indigenously Developed Surface to Surface missile with range of 150-500km

Warships : classification

- Various types
- Destroyers
- Frigates
- Aircraft carrier
- Corvette

Warships : classification

- Destroyers: Combat operation equipped with missiles and torpedoes
- Big sized Ships : Kolkata class, Rajput , Delhi
- Frigates; Small size, stealth, high maneuverability

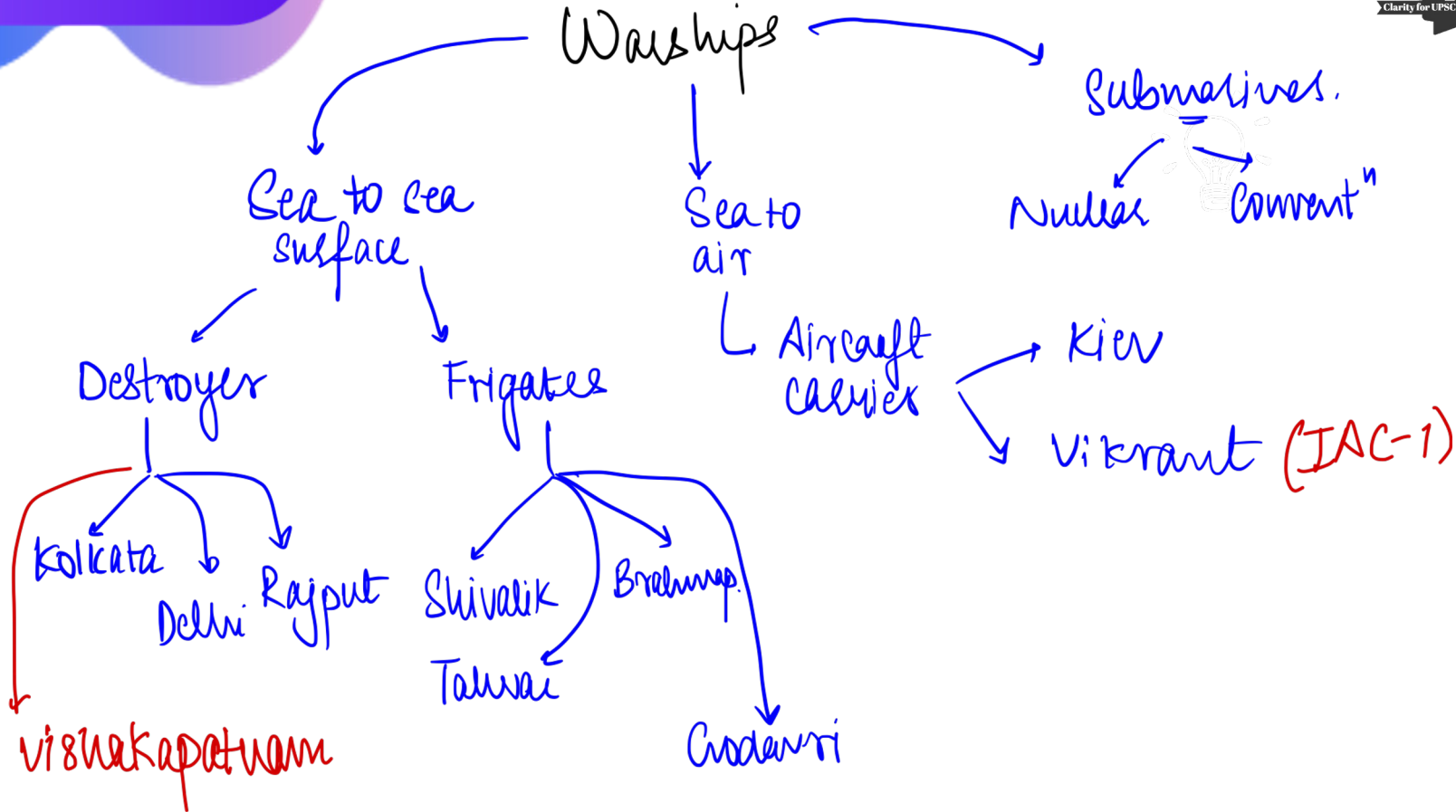
Warships : classification

- Aircraft carriers
- Biggest warships
- Runway and takeoff platform
- carry fighters and helicopter

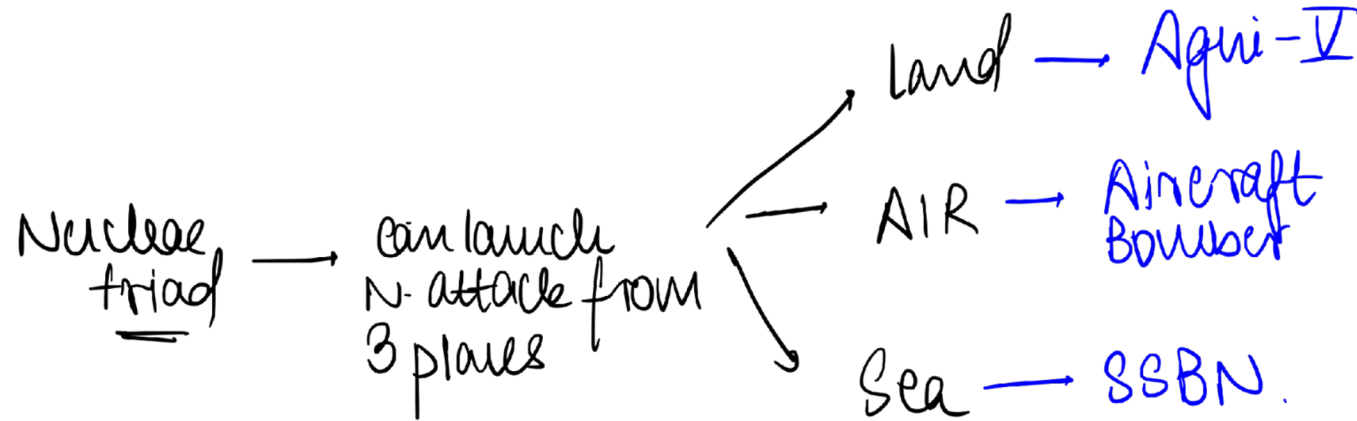
Warships : classification

Corvette

- Anti submarine warships (ASW) protect from submarine attack.



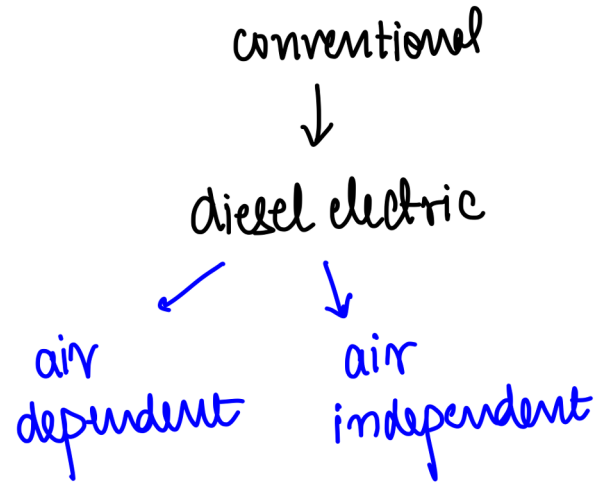
Understanding a Nuclear triad



- ① Second Strike Capability
- ② Mutually assured Destruction?

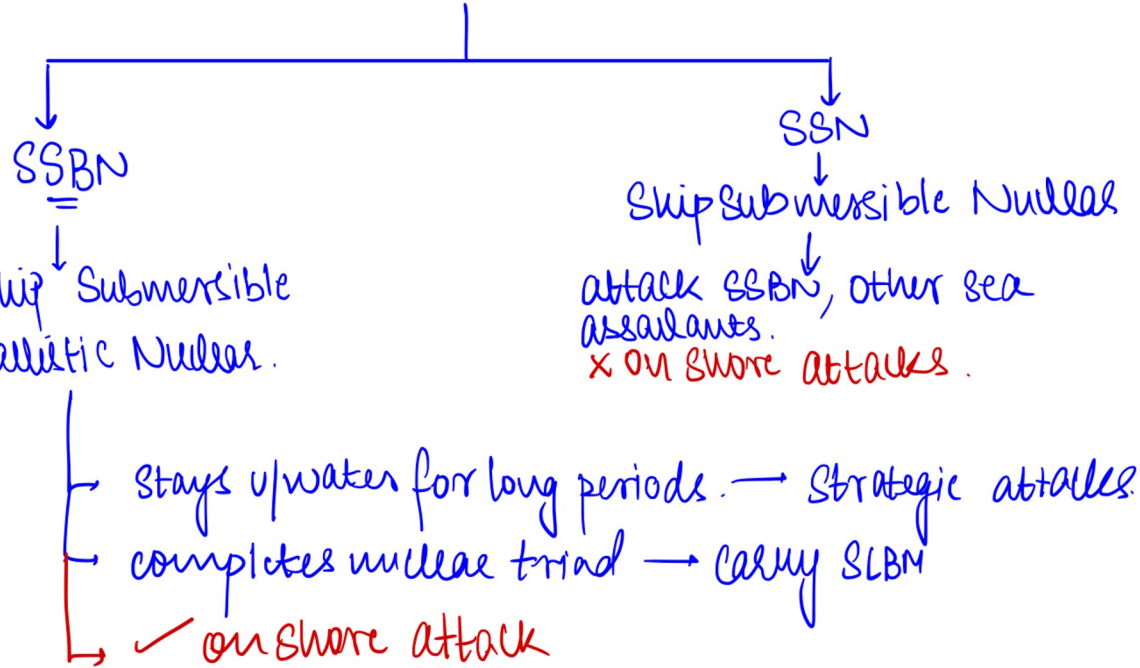
Submarines : types

- Conventional and Nuclear Submarine





Nuclear Submarine



Types : Based on capability

- An attack submarine is specifically designed for attacking and sinking other submarines, surface combatants and merchant vessels.
- They are also used to protect friendly surface combatants and missile submarines.
- Many modern attack submarines can launch cruise missiles (and dedicated anti-ship missiles) from their torpedo tubes, however main attack weapon remains the torpedo.

Ballistic Missile Subs

- A ballistic missile submarine is a submarine capable of deploying submarine-launched ballistic missiles (SLBMs) with nuclear warheads.
- The classification symbols for ballistic missile submarines are SSB and SSBN. The SS denotes submarine, the B denotes ballistic missile, and the N denotes that the submarine is nuclear powered.

Ballistic Missile Subs

- These submarines became a major weapon system in the Cold War because of their nuclear deterrence capability.
- They can fire missiles thousands of kilometers from their targets. Acoustic quieting makes them difficult to detect. Hence, they are a survivable deterrent in the event of a first strike and a key element of the mutual assured destruction policy of nuclear deterrence.

Ballistic Missile Subs

- The deployment of SSBNs is dominated by the United States and Russia. Smaller numbers are in service with France, the United Kingdom, China, and India.
- In 2009, India launched the first of its indigenously-built Arihant-class submarines.

Cruise missile submarine

- A cruise missile submarine is a submarine that carries and launches cruise missiles (SLCMs and anti-ship missiles) as its primary armament.
- The classification symbols for cruise missile submarines are SSG and SSGN. The SS denotes submarine, the G denotes guided missile, and the N denotes that the submarine is nuclear-powered.

Class of Submarine

- Vehicle class means the designation of motor vehicle types. It includes hatchback, sedans, sport utility vehicles, SUVs, trucks etc.
- A submarine class is a group of submarines of a similar design. Similarly, a ship class is a group of ships of a similar design.

Class of Submarine

- Submarines are built in classes, using a single design for a number of boats. Minor variations occur as improvements are incorporated into the design. However, in general, all boats of a class are noticeably similar.
- The 30 year's Indian Submarine Program (2000-2030) aims to build 24 submarines: 18 conventional and 6 nuclear powered (SSNs).

Indian Subs : Diesel electric submarines

- German subamrines : Shishumar class
- Russian Attack submarines : Sindhughosh class or kilo class
- Name of these submarines starts from Sindhu e.g. INS Sindhughosh (S55), INS Sindhudhvaj (S56), INS Sindhuraj (S57), INS Sindhuratna, (559), INS Sindhukesari (\$60), INS Sindhukirti (S61), INS Sindhuvijay (S62), INS Sindhurashtra (S65).

Indian Subs : Sindhu class



Indian Subs : Kalvari Class Scorpene

- Attack submarines, jointly developed by France and India.
- It has diesel propulsion and an additional air- independent propulsion (AIP) system
- Six submarines under this class: INS Kalvari, Khanderi, Karanj, Vela, Vagir, Vagsheer. INS Kalvari is also known as Tiger Shark (or S50).
- Developed under Project 75 or P 75 of the Indian Navy.
- Kalvari class of submarines design is a class of diesel-electric attack submarines.

Indian Subs : Scorpene and Kalvari



Nuclear Submarine

- It is a nuclear-powered ballistic missile submarine (SSBN). It can also launch missiles.
- INS Arihant (S2 or SSBN 80) is the lead ship of this class.
- INS Arighat (S3 or INS Aridhaman) is the second Arihant-class submarine.

Nuclear Submarine

INS ARIHANT



Akula Class

- Russian Powered Nuclear attack submarine : SSN
- INS chakra is an example



Vagir Submarine

- It is the fifth submarine (Kalvari, Khanderi, and Karanja, vela, vagsheer are other five) of the Project- 75 that was delivered to the Indian Navy recently.
- Project-75 includes construction of six submarines of Scorpene design.
- Scorpene submarines can undertake multifarious types of missions i.e Anti-Surface warfare, Anti-Submarine warfare, Intelligence gathering, Mine Laying, Area Surveillance etc.

Vagsheer Submarine

- Sixth and Last submarine of Project 75 has been launched.
- Diesel attack submarine.
- Status of P75 : kalvari, khanderi, karanj, vela have been commissioned.
- Vagir : Sea trials.

Project 75 I

- Project-75(I) envisages indigenous construction of six modern conventional Submarines with contemporary equipment, weapons & sensors including Fuel-Cell based AIP (Air Independent Propulsion Plant) etc.
- AIP technology allows conventional diesel-electric submarines to remain underwater for longer, enhancing its lethality.
- Other countries to have an AIP system include China, Germany, Sweden, France, Spain and Russia.
- P-75(I) submarine project is the first acquisition India has undertaken through its Strategic Partnership procurement model.

Project Sea Bird

- It involves creation of a naval base at Karwar (Karnataka) on the west coast of India.
- Upon completion, this will provide the Indian Navy with its largest naval base on the west coast and also the largest naval base east of The Suez Canal.

Types of Warships

- Aircraft Carrier :
- An aircraft carrier is a warship that serves as a dynamic seagoing airbase.
- It is equipped with a full-length flight deck.
- Typically, it is the capital ship of a naval fleet.

Types of Warships

- It allows a navy to project air power worldwide without depending on local air bases.
- By its diplomatic and tactical power, the aircraft carrier is the centerpiece of modern combat fleets.

Types of Ships

- Destroyer: It is a fast and maneuverable warship intended to defend larger vessels in a fleet.
- Frigates: it is a naval vessel intermediate between corvettes and destroyers.
- Corvette : Very small warship, fast and maneuverable
- Patrol vessels : Used to patrol coasts, example : ships of Indian Coast guard

Sea Denial and Sea Control

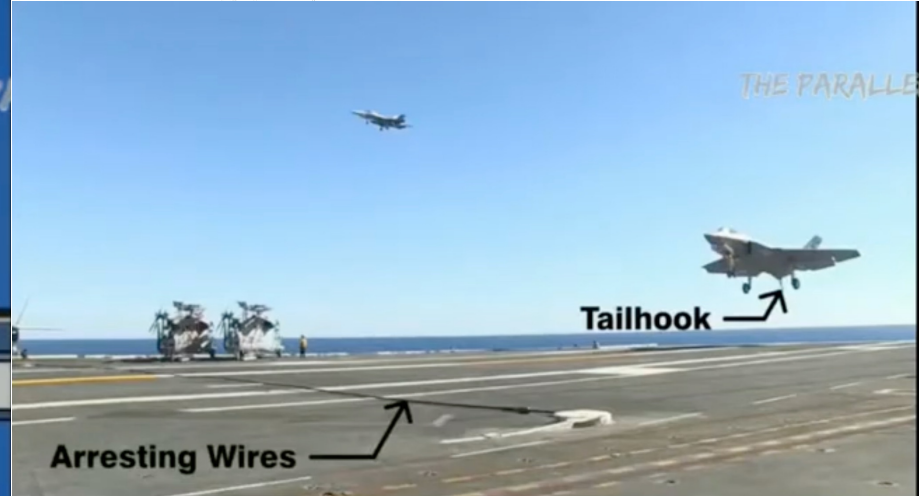
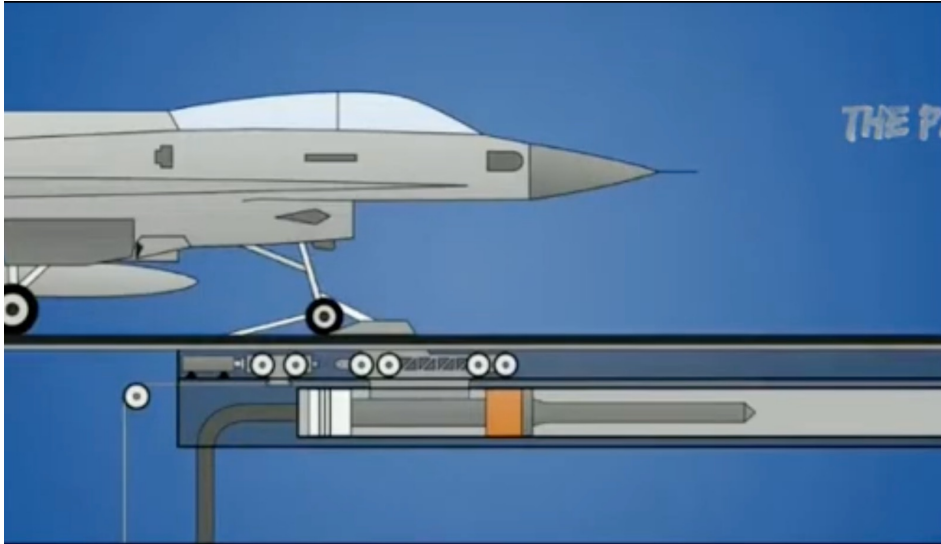
- Sea denial : word breakdown : no need of aircraft carrier
- Sea control : word breakdown : Needs aircraft carrier

Carriers

- STOBAR system (Short Take-Off but Arrested Recovery)
- A system used for the launch and recovery of aircraft from the deck of an aircraft carrier.
- Combines elements of both: Short take-off and vertical landing (STOVL) and Catapult- assisted take-off but arrested recovery (CATOBAR).

Carriers

- EMALS system (Electromagnetic Aircraft Launch System)
- The system launches aircraft by a catapult employing a linear induction motor rather than the conventional steam piston.





Indian Carriers

- INS Vikrant : majestic Class, Decommissioned in 1997
- INS virat : centaur class, Decommissioned in 2016
- INS Vikramaditya : KIEV CLASS, active aircraft carrier in India.
- INS Vikrant (IAC -1)

INS VIKRAMADITYA



Length
284 metre

Displacement
44,570 tonne

Height
60 metre
(20-storey tall, with 22 decks)

Max Speed
30 knots

Max Range | 7,000 nautical miles
at economical 18 knots speed.

CAPACITY (34 aircraft)

24

MIG-29Ks



10

Helicopters



Angled ski
jump for
fighters to
take off.



Arrestor gear for jets to hook on to
after touchdown on flight deck



Crew
1,600-1,800
Sailors

OPERATIONAL AIRCRAFT CARRIERS

US 11



2



2



China, UK, France,
Russia, Spain,
Brazil, Thailand
1 (each)

INS Vikrant- First Indigenous Aircraft Carrier (IAC)

- INS Vikrant is commissioned in 2022.
- Before this, we had a solitary aircraft carrier : INS vikramaditya
- Vikrant, a STOBAR (short take-off but assisted recovery) aircraft carrier, has an indigenous content of 76%.
- Designed by the Indian Navy's Directorate of Naval Design, Vikrant has been built at the state-owned Cochin Shipyard Limited.

INS Vikrant- First Indigenous Aircraft Carrier (IAC)

- Apart from India, only US, UK, Russia, France and China have the capability to build aircraft carriers.
- It will operate MiG-29K fighter aircraft, Kamov-31 Air Early Warning Helicopters, the soon-to-be-inducted MH-60R multi-role helicopter, and the indigenously manufactured Advanced Light Helicopters.
- It will provide the ability to project air power over long distances, airborne anti- submarine warfare and airborne early warning etc.



INS vishal (IAC 2)

- EMALS CATOBAR system.
- Nuclear propulsion.
- It Will be India's 2nd Indigenous Aircraft Carrier (IAC-2) to be built in India after INS Vikrant (IAC-1).



Think : where will India deploy its
carriers?

Destroyers

- Rajput Class : oldest
- Guided Missile Destroyer
- INS Rajput, INS Rana, INS Ranveer, INS Ranvijay.

- Delhi Class : Project 15 : GMD
- Indigenous
- Delhi, Mysore, Mumbai



Destroyers

- Kolkata Class : project 15A
- Stealth GMD
- The class comprises three ships - INS Kolkata, INS Kochi and INS Chennai.
- Barak – 8 capable
- Vishakhapatnam class : Project 15B
- Stealth GMD
- Brahmos and MRSAM capable



Y 12704 (Visakhapatnam)

- It is the lead ship of the Project-15B state-of-the-art stealth guided missile destroyers.
- It is equipped with BrahMos supersonic cruise missiles and medium range Surface- to-Air missiles.
- Contract for four ships of Project 15B was signed in 2011.

Y 12704 (Visakhapatnam)

- Four ships are christened after major cities viz. Visakhapatnam, Mormugao, Imphal and Surat.
- India's indigenous Destroyer construction programme commenced in the late 1990s with the three Delhi class (P-15 class) warships and this was followed by three Kolkata class (P-15A) destroyers commissioned a decade later.

Project 15B

- Mormugao and Surat ship of project 15B have also been launched recently
- Built at Mazagaon Dock Shipbuilders Limited
- Stealth Guided missile destroyers

Frigates

- Godavri Class : Project 16
- Guided Missile frigates : Indigenous
- Godavri, Ganga , Gomti

- Shivalik Class : Project 17
- First Stealth warship in India
- Stealth GMF

Frigates

- Project 17A : Taragiri and Udaygiri ships of Project 17A have been launched. They are advanced version of P17 frigates (Shivalik Class)
- Designed by : Indian Navy's warship design bureau (indigenous)
- 75% orders from indigenous firms including MSMEs.



Frigates

- Talwar Class : Russian
 - Stealth Guided Missile Frigates
 - Talwar, trishul, Tabar, teg, tarkash, trikhand
-
- Brahmaputra class
 - Guided missile frigate
 - Brahmaputra, Beas, Godavri



Corvette

- Anti Submarine Warfare
- Indigenous
- INS Kamorta, INS Kadmatt, INS Kilatan, INS Kavaratti
- INS kilatan : NBC warfare, Carbon fibre composite material
- INS Kavaratti : indigenous, NBC capable

Patrol Vessels

- Car nicobar class patrol vessel : water jet propelled
- Started after 2008 Mumbai attacks
- INS tarmugli : highly maneuverable Water Jet Fast Attack Craft (WJFAC).

Patrol Vessels

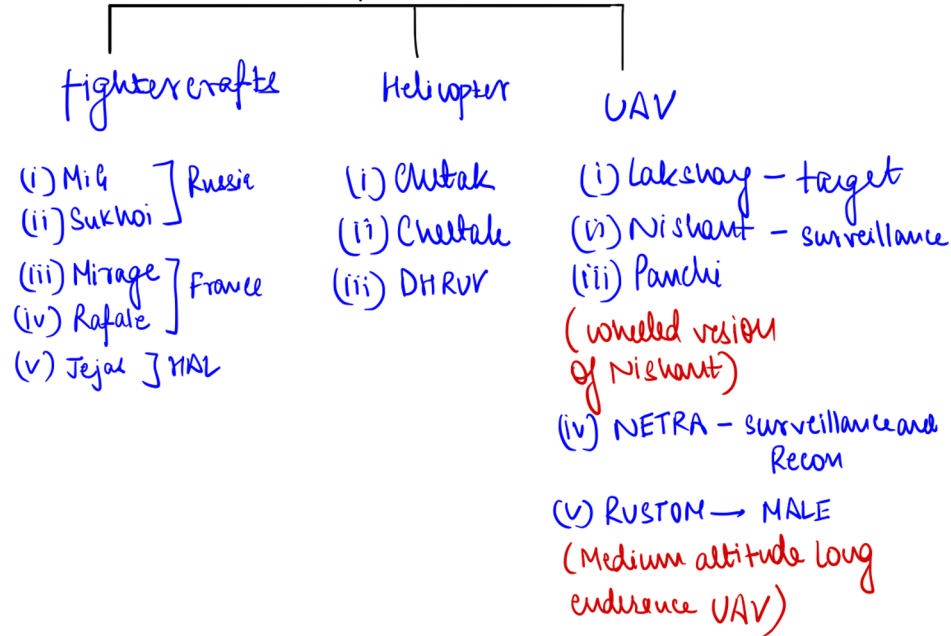
- Fast Patrol Vessel :
- Indian Coast guard, to control coasts
- Kankalata Barua, Priyadarshini , Annie Besant, Kamala Devi, Amrit Kaur

INS astradharini

- Torpedo launch and recovery
- Mine launch and recovery
- Indigenous



Air defense



Air Fleet

- Mig 21
- The Mikoyan-Gurevich MiG-21 is a Soviet designed supersonic jet fighter and interceptor.
- Interceptor: a fast aircraft for stopping or repelling hostile aircraft.



Air Fleet

- Rafale
- The Dassault Rafale is a French twin-engine, multirole, canard delta wing, fighter aircraft.
- Golden Arrows
- MMRCA : medium multi role combat aircraft



Rafale

Rafale advantages $\left\{ \begin{array}{l} \text{fast engine} \\ \text{air refuel} \\ \text{multimodal targeting} \end{array} \right.$

Air Fleet

- Sukhoi Su -30 Mkl
- It is a twin-engine, multirole, all-weather, long-range fighter.
- It is developed by Russia's Sukhoi and built by India's Hindustan Aeronautics Limited



Air Fleet

- Mirage 2000
- Dassault (French)
- Balakot airstrike



Generations of fighter jets

- 1st generation : piston based propellers, no avionics, manually controlled guns.
- 2nd generation : afterburner, infrared homing missiles
- 3rd generation : multirole, BVR, turbofan engine
- 4th generation : fly by wire, compound delta wing, hypermaneuverability
- 4.5 generation : use of composite materials to reduce radar cross section, better radars
- 5th generation : C3 : command control and communication and digital computation, better battlefield awareness. Ex : F-35, Chengdu J-20, Su-57
- 6th generation : in development

Tejas

- Indigenously developed lightweight multi role, Single engine, tail-less tactical fighter aircraft with compound delta wing. **4.5 generation.**

Special features.

- Compound delta platform
- Static stability
- Composite structure
- Fly-By-Wire flight control system

Meaning of these terms

- Fly by wire : Fly-by-wire (FBW) is a system that replaces the conventional manual flight controls of an aircraft with an electronic interface. The movements of flight controls are converted to electronic signals transmitted by wires, and flight control computers determine how to move the actuators at each control surface to provide the ordered response.

Meaning of these terms

- **Afterburner** : An afterburner (or reheat in British English) is an additional combustion component used on some jet engines, mostly those on military supersonic aircraft. Its purpose is to increase thrust, usually for supersonic flight, takeoff, and combat. The afterburning process injects additional fuel into a combustor in the jet pipe behind (i.e., "after") the turbine, "reheating" the exhaust gas. Afterburning significantly increases thrust as an alternative to using a bigger engine with its attendant weight penalty, but at the cost of increased fuel consumption (decreased fuel efficiency) which limits its use to short periods

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Relaxed Static Stability

relaxed stability refers to low or negative stability as a characteristic of aircraft.

An aircraft with negative stability will have a tendency to change its pitch and bank angles spontaneously.

Tejas

- Glass cockpit

It can carry A2A missile, A2G missiles, anti - ship missiles, laser guided bombs, drop tanks, Night targeting pod, electronic warfare system etc.

- Speed : 1.8 mach
- Kaveri Engine : GE 404 turbofan with afterburner available

Tejas MK2

Cabinet Committee on security has cleared tejas mk 2 which is a more capable version of indigenous tejas LCA.

4.5 generation aircraft with 70% indigenization.

GE 414 engine : longer range and greater weapon capacity than Tejas Mk-1

Other Aircraft

- Ilyushin IL-78 : Mid air refuel tanker
- Heron UAV : MALE Medium Altitude Long Endurance
- Chinook Helicopter : American made, received by India : used for HADR
- Apache Helicopter : Multi Role Heavy attack helicopter : Flying tank.



Aircraft in News

- Dornier 228 : The 19 seater HAL - DO - 228 aircraft is a highly versatile multi-purpose light transport aircraft. It has been developed specifically to meet the manifold requirements of utility and commuter transport, third level services and air-taxi operations, coast guard duties and maritime surveillance. It has been delivered to alliance air for civil operations in north east. Part of UDAN.
- LCH - Prachand : Indigenously developed and adopted by the Indian airforce. India has become the 7th country to make attack helicopters. Range : 550km, 6.5 km service ceiling (maximum altitude of flight). Only combat helicopter in the world that can take off and land at an altitude of 5000 meters.

Aircraft in News

- Indigenous stealth drone aka : Unmanned combat airborne vehicle has taken its maiden flight. UCAV is the precursor to the Ghatak armed stealth drone programmed called AURA : Autonomous Unmanned Research Aircraft.
- Chinook : US army has grounded chinook after finding that helicopter can be at risk of engine fires. India inducted chinook in 2019. Used for HADR.

Aircraft in News

- Kamikaze drones : suicide drones packed with explosives.
Also called switchblades. Army is inducting in eastern Ladakh.
They were used in Russia Ukraine war extensively.
- Hermes Starliner unmanned system : Israel has allowed
drones in civilian airspace (first country to do so)

Various types of detectors

(i) Hull Mounted sonar (HMSA)

- 3rd generation, indigenous, ship-borne, well mounted Sonar system designed and developed by DRDO.
- Active-cum passive integrated system, can be installed on various naval platforms. Active : sound sent by SONAR, Passive : Present signals are intercepted.
 - Frigates
 - Destroyers
 - ASW corvettes.

USHUS

Sonar for submarines with target classification facilities. It can be interfaced with onboard weapon systems.

Range > 30km

Various types of detectors

Dunking Sonar

- Advanced sonar used to survey under sea targets with the help of helicopter.
- The chopper is used for dunking sonar underwater, Range ~ 1 km

Various types of detectors

Battlefield Surveillance Radar

- Man portable battery operated surveillance radar. Simple, easy to use and can be installed anywhere in the battlefield.
- Audio and visual aids help identify and classify target.

Various types of detectors

Rohini

- Indigenous, stand alone, medium Range, 3D surveillance radar for air force. It can scan the airspace and use multi-beams technology to identify targets accurately.
- Range = 200 km.

Various types of detectors

Bharani

- Low level, Lightweight, L-Band, 2D radar used in mountainous terrain against hostile arial target (UAV, chopper, Low and medium altitude aircraft)

Various types of detectors

Revathi

- 1st Indigenous Radar and also 1st in Navy to use digital Infrared frequency tech.
- Radar can auto trace upto 150 targets with Range = 200 km.

Various types of detectors

Ashlesha Radar : ground based, S band, 3D, low level leight weight radar with range of 60km.

Swathi Radar : Weapon locating radar, automatically locates hostile artillery, mortars and rocket launchers by using signal processing technique.

CHIEF OF DEFENCE STAFF (CDS)



Chief of Defence Staff (CDS)



About: Approved in 2019, is the **head of Army, Navy and Air Force** and is a **four-star military officer**.



Eligibility: Any **serving or retired Lieutenant General, Air Marshal and Vice Admiral** under age of 62 years.



Proposed by: Kargil Review Committee (2000), **Naresh Chandra Committee** on defence and security (2011) and **Shekatkar Committee** (2016).



Tenure: Central government may extend tenure to **maximum of 65 years**.



Roles and powers of CDS

- ◆ **Heads Department of Military Affairs** within Ministry of Defence and **acts as its secretary**.
- ◆ **Permanent chairman** of existing **Chiefs of Staff Committee**.
- ◆ **Member of Defence Acquisition Council** and **Defence Planning Committee**.
- ◆ **Principal military advisor** to **Defence Minister** and **military advisor** to **Nuclear Command Authority**.



GSAT 7 SERIES SATELLITES

- GSAT 7 series satellites are advanced satellites developed by ISRO to meet the communication needs of the defence services.
 - GSAT 7 satellite : used by the Indian Navy
 - GSAT 7A : boosting the connectivity between the ground radar stations, airbases and the airborne early warning and control aircraft (AEW&C) of the IAF (Indian Air Force).
- GSAT 7B : needs of the Army.

Operation 'Skylight'

- It involved the army testing the readiness of its satellite based systems.
- Unlike the Indian Air Force and the Navy, the army currently does not have a dedicated satellite.
 - Currently, the Army is using 30% of the communication capabilities of the GSAT 7A.
 - Defence Acquisition Council has cleared an army proposal for a GSAT-7B satellite.

Other news

- Dirty bomb : non highly refined nuclear material like uranium obtained from hospitals, nuclear power stations or research labs. Cheaper, quicker to make.
- Sweden has announced plans to manufacture Carl Gustaf M4 in India. It is a man portable recoil less rifle with ability to fire variety of ammunition.
- Pinaka Multi Barrel Rocket Launcher : Armenia has placed first ever export order of India's MBRL, indigenous rocket system developed with DRDO. It is a mobile rocket launch system with range 40km (mk 1) to 75km (Pinaka ER : extended range)

Other news

- Submersible platform for acoustic characterization and evaluation : testing and evaluation facility for SONAR systems by DRDO.
- Abhyas : High speed expendable aerial target developed by DRDO.
- High Energy Laser Systems : SFO technologies is partnering with DRDO to develop HELS. They emit thermal energy to directly render a target ineffective without using a conventional munition. It is a type of Directed Energy weapon. India has launched various projects for HELS : Durga, KALI, Aditya.

Q.1 PYQ (2014)

Q. Which reference to Agni-IV Missile, which of the following statements is/are correct?

1. It is a surface-to-surface missile.
2. It is fuelled by liquid propellant only.
3. It can deliver one-tonne nuclear warheads about 7500 km away.

Select the correct answer using the code given below.

- a) 1 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2 and 3

Q.2 PYQ (2016)

Q. Which one of the following is the best description of ‘INS Astradharini’, that was in the news recently?

- a) Amphibious warfare ship
- b) Nuclear-powered submarine
- c) Torpedo launch and recovery vessel
- d) Nuclear-powered aircraft carrier

Q.3 PYQ (2022)

Q. Orbital Bombardment System” often talked about in media?

- (a) A hypersonic missile is launched into space to counter the asteroid approaching the Earth and explode it in space.
- (b) A spacecraft lands on another planet after making several orbital motions.
- (c) A missile is put into a stable orbit around the Earth and deorbits over a target on the Earth.
- (d) A spacecraft moves along a comet with the same surface. speed and places a probe on its

Q4) PYQ (2018)

Q) What is "Terminal High Altitude Area Defense (THAAD)", sometimes seen in the news?

- a) An Israeli radar system
- b) India's indigenous anti-missile programme
- c) An American anti-missile system
- d) A defence collaboration between Japan and South Korea

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c) 1 and 3 only

d) 1, 2 and 3

Answer: A

- Agni-IV is the fourth in the Agni series of missiles which were earlier known as Agni II prime.
- The missile is light-weight and has two stages of solid propulsion and a payload with re-entry heat shield.
- The strategic **surface-to-surface missile** was flight tested from launch complex-4 of the Integrated Test Range (ITR) at Dr Abdul Kalam Island, earlier known as Wheeler Island,
- The indigenously developed sophisticated **Agni-IV having 4,000-km strike** range is a two-stage missile

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- d) Nuclear-powered aircraft carrier

Answer: C

- INS Astradharini is an indigenously designed and **built torpedo launch and recovery vessel built by Shoft Shipyard for the Indian Navy.**
- She was commissioned to Navy Service on 6 October 2015 at the Naval Base in Visakhapatnam.
- After commissioning, the ship entered the Eastern Naval Command.

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Answer: C

- In 1960s, the Fractional Orbital Bombardment System (FOBS) was developed by the Soviet Union, an orbital bombardment system that could carry a nuclear weapon into orbit at a lower trajectory than a traditional fixed trajectory ballistic missile.
- The idea behind FOBS is, a warhead is put into a stable orbit and it deorbits over the target.
- If the target and the launch position is lined up and the warhead keeps going round, it will complete a circle.
- The real point is that missile can be flown directly to the target, or it is fired in the opposite direction the other way around the planet and still arrive at the target from the opposite direction.

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Answer: C

- Terminal High Altitude Area Defense (THAAD), formerly Theater High Altitude Area Defense is an American anti-ballistic missile defence system designed to shoot down short, medium, and intermediate-range ballistic missiles.
- It is used by USA